

23/12/22

CST Third Semester Final Examination, December 2022
Signals and Systems (CS-2104)

Answer any five questions.

Full Marks 50

Time 180 minutes

1. A signal is described by $x(t) = 2\cos(2t + 1) - \sin(5t - 1)$.
What is the periodicity of this signal? Given a long sequence $x[n]$ suggest some method to find out the time period of the sequence, if any. Explain the working of your methodology with small examples. [4+3+3]
2. An opamp based integrator system is designed using $R = 10K\Omega$ and $C = 0.5\mu F$. What will be the output signal if the input signal is a square wave with peak-to-peak amplitude of $1V$ and frequency of $2KHz$? How does the output signal nature change when the input signal frequency is changed to $500KHz$? Explain the filtering properties of the designed integrator system. [3+3+4]
3. Describe amplitude modulation and demodulation techniques for communicating an audio signal bandlimited to $15KHz$ over a distance range within $25Km$. What are the disadvantages of amplitude modulation? Suggest some mechanism to overcome these disadvantages. [4+2+4]
4. Consider a second order system described by the differential equation $\ddot{y} + 7\dot{y} + 12y = u(t)$ with $y(t)$ as output and $u(t)$ as input of the system. Obtain the transfer function of the system.
Derive the state transition matrix for this system by choosing two state variables one derivative apart. [4+6]
5. Obtain the Laplace transform of a rectangular pulse train. What happens when the pulse width becomes very small? Express in suitable transform domain the effect on a signal when such pulse train is used to modulate the signal. [3+3+4]
6. A spring mass system has spring constant K and damping factor f . Describe the system model in terms of a second order differential equation with natural frequency ω_n and damping coefficient ζ . In terms of this ω_n and ζ what is the response of the system to a unit step input signal? From the transfer function of the system, give an idea about the frequency response $H(j\omega)$. [3+3+4]
7. Write short notes on any two: [5+5]
 - (a) Design of PID controller circuits using opamps
 - (b) Time and frequency division multiplexing in communication systems
 - (c) Impulse response of linear time invariant systems.
8. Explain the following mathematically (any two): [5+5]
 - (a) System stability can be achieved through negative state feedback.
 - (b) The left half of complex s-plane maps to the unit circle of z-plane.
 - (c) Square wave signal has a duality with an ideal lowpass filter.